ModelIt!

Teacher's Guide

Can an Algorithm Be Racist?

Modeling Bias in Artificial Intelligence and Machine Learning

Quick Reference

Grade Level	Grade 11L1
NGSS Standard	HS-ETS1-1, HS-ETS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ETS1-1, HS-ETS1-4

Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use computational simulations to model how training data composition and algorithm design choices produce measurably different outcomes for different demographic groups.
Disciplinary Core Idea	**ETS1.A Defining and Delimiting Engineering Problems / ETS1.B Developing Possible Solutions** — Algorithmic fairness is an engineering design problem that requires specifying quantitative criteria (disparate impact ratio, error rate parity) and constraints (accuracy, cost, legal compliance) while accounting for societal needs and values.
Crosscutting Concept	**Cause and Effect** — Students trace the causal pathway from historical discrimination through biased training data to disparate algorithmic outcomes, identifying how proxy variables transmit racial information even when race is not explicitly used.

CAST Testing Alignment

MULTIPLE CHOICE:

A lending algorithm trained on historical data denies loan applications from predominantly Black zip codes at twice the rate of predominantly white zip codes, even though it does not use race as an input variable. What is the most likely explanation? A) The algorithm is functioning correctly, B) Zip code is serving as a proxy variable for race, C) The algorithm has a programming error, D) Black applicants are genuinely less creditworthy

CONSTRUCTED RESPONSE:

Using your model, explain why removing race from an algorithm's feature set does not eliminate racial bias. Describe the role of proxy variables and provide a specific example from your simulation showing how bias persists after removing demographic features.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

In 2016, ProPublica investigated COMPAS, an algorithm used across the United States to predict whether criminal defendants would reoffend. They found that the algorithm was twice as likely to falsely label Black defendants as high-risk compared to white defendants. The algorithm never used race as an input variable. This investigation revealed a fundamental truth about artificial intelligence: algorithms are not objective — they are mirrors that reflect the biases embedded in the data they learn from.

How Machine Learning Learns Bias

Machine learning algorithms find patterns in data. If the historical data contains discriminatory patterns — and virtually all real-world data does — the algorithm will learn and replicate those patterns. A hiring algorithm trained on a tech company's 20-year hiring history will learn that successful hires tend to be white and male, not because those demographics are more capable, but because historical hiring decisions were biased. The algorithm cannot distinguish between genuine performance predictors and artifacts of discrimination.

The Proxy Variable Problem

Even when developers remove explicit demographic variables (race, gender, ethnicity) from an algorithm's input features, bias persists through proxy variables — seemingly neutral data points that are strongly correlated with protected characteristics due to historical patterns. Zip code predicts race because of residential segregation caused by redlining. School name predicts socioeconomic status. Arrest history predicts race because of discriminatory policing. Credit score predicts race because of disparities in lending and employment. An algorithm using these proxies can produce racially discriminatory outcomes while appearing race-blind.

Real-World Algorithmic Harm

Algorithmic bias is not theoretical — it is causing measurable harm today. Amazon scrapped an AI hiring tool in 2018 after discovering it systematically downgraded resumes containing the word 'women's.' Healthcare algorithms used by hospitals across the US were found to systematically underestimate the health needs of Black patients, reducing their access to care. Facial recognition systems have error rates 10-100x higher for dark-skinned women than light-skinned men. Predictive policing algorithms send more officers to Black neighborhoods, generating more arrests that 'confirm' the algorithm's predictions in a vicious feedback loop.

Toward Algorithmic Fairness

Researchers have developed multiple mathematical definitions of fairness, including demographic parity (equal selection rates), equalized odds (equal error rates), and calibration (equal accuracy of risk scores). The challenge is that these definitions are mathematically incompatible — satisfying one often violates another. This means algorithmic fairness requires VALUE JUDGMENTS about which type of fairness matters most in each context. Technical solutions include debiasing training data, adding fairness constraints during model training, and post-processing outputs to equalize metrics. But ultimately, ensuring algorithms are fair requires governance: mandatory bias audits, impact assessments, transparency requirements, and meaningful human oversight of automated decisions that affect people's lives.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify training data composition, feature selection, algorithm fairness constraint, prediction accuracy, and disparate impact ratio as the key components of the algorithmic bias system.

E **Establish Relationships**

Students determine that training data composition, feature selection, and fairness constraints are external design choices made by developers, while prediction accuracy and disparate impact ratio are outcomes determined by the interaction of data, algorithm design, and the underlying patterns in the data.

V **Visualize & Model**

Students build a computational model that simulates how biased training data flows through a machine learning pipeline to produce measurably different outcomes for different demographic groups.

E **Evaluate Outcomes**

Students run biased historical data, debiased training data, and fairness-constrained algorithm scenarios to quantify how different interventions affect both accuracy and fairness.

R **Revise & Extend**

Students add feedback loop amplification, transparency level, or stakeholder impact to explore how bias compounds over time and how the consequences vary across different decision domains.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"An algorithm decides whether you get a loan, a job, or bail from jail. It doesn't know your race. But it knows your zip code, your school, and your credit score — and in America, those tell the same story."*

Watch For

Show the ProPublica COMPAS investigation headline. Ask: How can an algorithm that never sees your race still discriminate based on race?

Slide 2: Learning Objectives

3 min

Say: *"Today we're putting AI on trial. We're going to build a model that shows exactly how bias gets into algorithms and exactly what it takes to get it out."*

Watch For

Have students read objectives. Pre-teach 'proxy variable' and 'disparate impact' with concrete examples (zip code as proxy for race).

Slide 3: Big Question

3 min

Say: *"If an algorithm is just doing math on historical data — and the history is racist — who is responsible for the racist outcome? The data? The programmer? The company? Society?"*

Watch For

Quick-write: Students write who they think is responsible when an algorithm produces discriminatory outcomes. Discussion: Can math be racist?

Slide 4: LEVER Framework

2 min

Say: *"We're going to train an algorithm, measure its bias, try to fix it, and discover the trade-offs involved in making AI fair."*

Watch For

Preview LEVER steps. Emphasize that this is about measuring bias with data, not just arguing about it.

Slide 5: Activity 1: Components

8 min

Say: *"What determines whether an algorithm is fair? Some factors are choices developers make, others are measurable outcomes."*

Watch For

Guide sorting: Training Data Composition, Feature Selection, and Fairness Constraints are external (developer choices). Prediction Accuracy and Disparate Impact Ratio are internal (measurable outcomes).

Slide 6: Activity 2: Connections

8 min

Say: *"Biased data in, biased decisions out. But the proxy variable connection is the sneakiest: you can remove race from the data and the algorithm STILL discriminates. Let's trace how."*

Watch For

Students map the path from historical discrimination through training data to proxy variables to disparate impact. Key insight: removing race doesn't remove racism from the data.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's run three versions of the same hiring algorithm: the default trained on biased data, one with debiased data, and one with a fairness constraint. Which one is fair? Which one is accurate? Can you have both?"*

Watch For

Students predict accuracy and fairness metrics for each configuration. Run simulations. Analyze the accuracy-fairness trade-off.

Slide 8: Discoveries

5 min

Say: *"The model proves two things: first, algorithms CAN be racist even without using race. Second, we CAN make them fairer — but it requires deliberate effort and value judgments that math alone can't provide."*

Watch For

Discuss: If the trade-off between accuracy and fairness is small, why haven't all companies fixed their algorithms? What incentives work against fairness?

Slide 9: STEM Challenge

12 min

Say: *"A tech company's AI hiring system is discriminating by zip code. You've been hired to audit it, find the bias, and fix it. Your evidence has to be so strong it would hold up in court."*

Watch For

Groups design audit protocols, test hypotheses about proxy variables, and propose corrections with quantified fairness improvements. Present findings.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Training Data Composition, Feature Selection, Algorithm Fairness Constraint
- **INTERNAL:** Prediction Accuracy, Disparate Impact Ratio

Relationships

- **Training Data Composition to Disparate Impact Ratio:** DETERMINES — If training data reflects historical discrimination — where certain demographic groups were systematically disadvantaged — the algorithm learns those patterns as 'normal' and reproduces them. Biased training data directly produces low Disparate Impact Ratios (discriminatory outcomes) regardless of the algorithm's design.
- **Feature Selection to Disparate Impact Ratio:** MODIFIES — Including proxy variables (zip code, school name, credit score) that correlate with race allows the algorithm to discriminate even without explicit demographic features. Removing or carefully engineering features can reduce but not eliminate bias if the underlying training data is biased.
- **Algorithm Fairness Constraint to Prediction Accuracy:** NEGATIVE (-) SLIGHT — Imposing fairness constraints (e.g., requiring equal false positive rates across groups) typically reduces overall prediction accuracy by a small but measurable amount, because it forces the algorithm to sacrifice some predictive power to achieve equitable outcomes. Research shows this trade-off is usually modest (1-3% accuracy reduction) but not zero.

Key Discoveries

- Algorithms trained on biased historical data reproduce and amplify existing discrimination — high overall accuracy can mask systematic bias against specific demographic groups because the majority group dominates the accuracy metric
- Removing explicit demographic variables (race, gender) from the algorithm does NOT eliminate bias because proxy variables (zip code, school name, arrest history) carry the same information due to historical segregation and discrimination
- Debiasing training data significantly improves fairness (higher Disparate Impact Ratio) with minimal loss of overall prediction accuracy — proving that much of the bias was noise, not signal
- Fairness constraints create a real but modest trade-off with overall accuracy — society must decide whether a small decrease in overall accuracy is an acceptable price for preventing algorithmic discrimination in life-changing decisions

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Algorithms are inherently objective because they use math, not feelings"	Algorithms are built by humans, trained on human data, and optimized for human-defined objectives. Every step involves human choices: what data to collect, what features to include, what outcome to predict, what metric to optimize. These choices embed values and biases. The math itself may be neutral, but the inputs and objectives are profoundly human. An algorithm trained on biased data will produce biased outputs with mathematical precision — it is objectively implementing subjective bias.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If the algorithm is accurate overall, it must be fair"	High overall accuracy can mask severe unfairness for minority groups. An algorithm with 85% overall accuracy might have 90% accuracy for the majority group and 60% accuracy for the minority group — because the majority group dominates the overall metric. This is why disaggregated metrics are essential: you must measure accuracy and error rates separately for each demographic group to detect disparate impact. The model clearly shows that 'accurate on average' does not mean 'fair for everyone.'
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Just remove race from the data and the algorithm will be fair"	This is perhaps the most dangerous misconception because it feels so intuitive. Removing race from the input features changes almost nothing about the algorithm's outputs because proxy variables — zip code, school district, credit score, arrest record — carry racial information through historical correlations. In a segregated society, race is encoded in nearly every socioeconomic variable. True fairness requires active intervention: auditing outcomes by demographic, testing for proxy effects, imposing fairness constraints, and ongoing monitoring.